

# THANUSHYA VENUGOBAN

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[github.com/Thanu-Venu](https://github.com/Thanu-Venu) | [portfolio-thanu-venu.vercel.app](https://portfolio-thanu-venu.vercel.app)

## Professional Summary

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Computer Science undergraduate at the University of Colombo School of Computing (UCSC) with hands-on, end-to-end experience across full-stack web and mobile development, distributed and event-driven backend systems, DevOps/CI-CD pipelines, and AI-integrated applications. Comfortable owning a project from requirements through deployment — REST/WebSocket API design, relational and NoSQL data modeling, automated testing, and monitoring — in both independent and Agile team settings. Has led a 9-month, 4-member academic project through its full development lifecycle and independently built and deployed several production-style applications. Seeking a Software Engineering internship to apply this range on real, production systems.

## Technical Skills

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**Languages:** Python, JavaScript, TypeScript, PHP, Kotlin, C, C++, SQL, HTML, CSS

**Web & Mobile:** React, Node.js, FastAPI, Tailwind CSS, Jetpack Compose, REST API Design, WebSockets, JWT Authentication, MVC Architecture

**Databases:** PostgreSQL, MySQL, MongoDB, Firebase Realtime Database

**DevOps & Testing:** Docker, Docker Compose, GitHub Actions CI/CD, RabbitMQ, Prometheus, Grafana, Jest & Supertest, Git & GitHub

**AI & Cloud:** Gemini API (Generative AI Integration), Cloud Deployment (Render, Vercel, Firebase)

**Collaboration:** Agile/Scrum, Jira, Figma

## Projects

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### SmartCare — Caretaker Management System (2025 – 2026) *PHP, MySQL, JavaScript, MVC Architecture*

- Led development (largest contributor by commit share) of a 9-month, 4-member academic group project: a layered MVC PHP platform serving four roles (Admin, HR, Caretaker, Client) with role-based dashboards and access control.
- Built a recurring billing engine (hourly/daily/monthly/yearly plans) with advance payments, automated reminders, grace periods, and cron-driven auto-cancellation, integrated with the PayHere payment gateway; designed a versioned MySQL schema evolved through migrations.
- Implemented booking reassignment/reschedule and refund workflows, plus leave and attendance tracking; collaborated via Git/GitHub, Jira, and Figma in an Agile team workflow.

**Repository:** <https://github.com/Thanu-Venu/caretaker-management-system>

### DevFlow — Full-Stack Task Management Platform (2026) *React, Node.js, MongoDB, Docker, GitHub Actions*

- Built JWT-authenticated, per-user REST APIs backed by an automated Jest/Supertest test suite run in CI.
- Set up a GitHub Actions CI/CD pipeline (lint, test, Docker build, deploy) with independent frontend/backend releases to Render.
- Containerized the stack with Docker Compose and instrumented it with Prometheus metrics and Grafana dashboards for application monitoring.

**Live Demo:** <https://devflow-frontend-gufn.onrender.com>

**Repository:** <https://github.com/Thanu-Venu/devflow>

### SwiftLogistics — Distributed Logistics Middleware Platform (2026) *FastAPI, React, Docker, PostgreSQL, RabbitMQ*

- Designed and built an event-driven middleware platform (API Gateway, worker orchestrator, RabbitMQ) integrating three heterogeneous backend systems, each over a different protocol/format: CMS (SOAP/XML), ROS (REST/JSON), and WMS (TCP).
- Exposed REST and WebSocket APIs for real-time order tracking through a React client portal; applied a retry/dead-letter-queue pattern for fault tolerance.
- Containerized the multi-service architecture with Docker Compose for consistent, repeatable deployment.

**Repository:** [https://github.com/Thanu-Venu/SwiftLogistics\\_Middleware](https://github.com/Thanu-Venu/SwiftLogistics_Middleware)

## Smart Home Monitoring & Control System (2026) *Kotlin, Jetpack Compose, React, Firebase*

- Built a real-time smart-home platform with a native Android app (Kotlin, Jetpack Compose, MVVM) and a React web simulator sharing a single Firebase Realtime Database, so device changes sync live across both clients.
- Implemented per-device/per-switch schedule automation and a safety-cutoff system that auto-disables a device past a configurable ON duration.
- Collaborated in a 3-member team on the shared data model, Firebase security rules, and real-time sync logic across both clients.

Repository: <https://github.com/Thanu-Venu/smart-home-monitoring-system>

## Bank Turnover Analyzer (2026) *Python, FastAPI, PostgreSQL, React, Gmail API*

- Built an automated pipeline that fetches bank e-statements via the Gmail API, parses transactions from PDF statements, and loads them into PostgreSQL for reporting.
- Built a FastAPI backend and React dashboard for monthly/yearly reporting, backed by dedicated audit, deduplication, and ownership-assignment scripts.

Repository: <https://github.com/Thanu-Venu/bank-turnover-analysis>

## AI-Powered Developer Portfolio (2026) *React, FastAPI, Gemini API, Vercel, Render*

- Built and deployed this portfolio site with a custom AI chatbot assistant powered by the Gemini API, backed by a FastAPI service.
- Implemented per-IP rate limiting and response caching on the backend to control latency and API cost under public traffic.

Live Demo: <https://portfolio-thanu-venu.vercel.app/>

## Publish-Subscribe Middleware (2026) *Python, TCP Sockets, CLI*

- Implemented a publish-subscribe middleware from raw TCP sockets in a 4-member team, supporting concurrent publisher/subscriber clients and topic-based message filtering.
- Iteratively extended the architecture across course milestones, from basic client-server messaging to asynchronous topic-based distribution.

Repository: [https://github.com/Thanu-Venu/pub\\_sub\\_assignment1](https://github.com/Thanu-Venu/pub_sub_assignment1)

## Additional Projects

Mini Compiler — C, lexical analysis & recursive-descent parsing — [github.com/Thanu-Venu/mini-compiler-c](https://github.com/Thanu-Venu/mini-compiler-c)  
Ludo Game Simulation — C, turn-based game engine with structs & arrays — [github.com/Thanu-Venu/LUDO-LIKE-UCSC](https://github.com/Thanu-Venu/LUDO-LIKE-UCSC)

## Education

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|----------------------------------------------------------------------------|--------------------|
| University of Colombo School of Computing (UCSC), Sri Lanka                | May 2024 – Present |
| BSc (Hons) in Computer Science, GPA: 3.594/4.00                            |                    |
| V/Rambaikulam Girls' Maha Vidyalayam, Vavuniya, Sri Lanka                  | 2010 – 2023        |
| GCE A/L — Combined Mathematics, Physics, Chemistry (AAB)   Z-score: 1.8869 |                    |

## Certifications

Web Design for Beginners — University of Moratuwa | HackerRank — Problem Solving & Programming

## References

Ms. S. Hamsavasini — Lecturer, UCSC; Project Supervisor, SmartCare | +94 77 617 6728 | [vhs@ucsc.cmb.ac.lk](mailto:vhs@ucsc.cmb.ac.lk)  
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